CSE 202: Object Oriented Programming in C++ — Unit 1

Multiple Choice Questions Practice Sheet (Shuffled Order)

1. Which of the following is NOT a fundamental pillar/feature of Object-Oriented Programming?

A. Encapsulation

B. Inheritance

C. Polymorphism

D. Compilation

Correct Answer: D. Compilation

2. By default, members of a C++ `class` are:

A. Public

B. Private

C. Protected

D. Static

Correct Answer: B. Private

3. Which operator is used with `cin` for taking input in C++?

A. `<<` (Insertion operator)

B. `>>` (Extraction operator)

C. `->`

D. `::`

Correct Answer: B. `>>` (Extraction operator)

4. Which keyword is used to define a class in C++?

A. `object`

B. `class`

C. `define`

D. `structure`

Correct Answer: B. `class`

5. Which statement correctly creates an object `s` of a class named `Student`?

A. `Student : s;`

B. `new Student s;`

C. `Student s;`

D. `s Student;`

Correct Answer: C. `Student s;`

6. What will be the output of the following code snippet?

```
class Test {
public:
    int x = 20;
};

int main() {
    Test t;
    cout << t.x;
    return 0;
}
```

A. 0

B. 20

C. Garbage value

D. Compilation Error

Correct Answer: B. 20

7. Which access specifier makes class members accessible from outside the class?

A. `private`

B. `protected`

C. `public`

D. `friend`

Correct Answer: C. `public`

8. Which feature allows multiple functions in the same scope to share the same name but have different parameter lists?

A. Function Overriding

B. Function Overloading

C. Function Recursion

D. Inline Expansion

Correct Answer: B. Function Overloading

9. What is the correct rule regarding default arguments in C++ functions?

A. Default arguments must always be the first parameters

B. Default arguments must always be trailing (rightmost) parameters

C. Every parameter in a function must have a default value

D. Default arguments cannot be used with overloaded functions

Correct Answer: B. Default arguments must always be trailing (rightmost) parameters

10. What is an inline function in C++?

A. A function that cannot return any value

B. A function whose code may be expanded inline at the point of function call to save call overhead

C. A function that executes only once per program run

D. A function that must always be recursive

Correct Answer: B. A function whose code may be expanded inline at the point of function call to save call overhead

11. Which keyword is traditionally used to declare an inline function?

A. `inline`

B. `inside`

C. `expand`

D. `fast`

Correct Answer: A. `inline`

12. Which of the following is a standard C++ I/O manipulator used for creating a new line?

A. `endl`

B. `cin`

C. `cout`

D. `class`

Correct Answer: A. `endl`

13. Which header file is required to use parameterized formatting manipulators such as `setw` or `setprecision`?

A. ``

B. ``

C. ``

D. ``

Correct Answer: B. ``

14. What is a static data member of a class?

A. A member that belongs separately to each individual object

B. A member shared by all objects of the class (single copy)

C. A member that cannot be modified anywhere

D. A member that exists only inside local function blocks

Correct Answer: B. A member shared by all objects of the class (single copy)

15. How is a static member function `show()` belonging to class `Test` generally called from `main()` without an object instance?

A. `Test::show();`

B. `Test.show();`

C. `Test->show();`

D. `Test show();`

Correct Answer: A. `Test::show();`

16. Which statement about a static member function is TRUE?

A. It can directly access non-static data members without an object

B. It strictly requires an object instance to be invoked

C. It can directly access only static data members and other static functions

D. It cannot access any class members at all

Correct Answer: C. It can directly access only static data members and other static functions

17. What is the main difference between Procedural Programming and Object-Oriented Programming (OOP)?

A. Procedural focuses primarily on functions/procedures, while OOP focuses on objects containing data and methods

B. Procedural programming does not use variables

C. OOP cannot use functions

D. There is no structural difference

> **Correct Answer:** A. Procedural focuses primarily on functions/procedures, while OOP focuses on objects containing data and methods

18. Which of the following is a classic example of Encapsulation?

A. Binding data and functions operating on that data together inside a single class unit

B. Calling a function recursively

C. Using a `for` loop to iterate arrays

D. Using `cout` for printing output

> **Correct Answer:** A. Binding data and functions operating on that data together inside a single class unit

19. What is a friend function in C++?

A. A function that is automatically inherited by derived classes

B. A non-member function granted access to `private` and `protected` members of a class

C. A function that must always be a member of the class

D. A function that cannot accept any parameters

Correct Answer: B. A non-member function granted access to `private` and `protected` members of a class

20. Which statement about a friend function is TRUE?

A. It is invoked using object dot operator (`object.friendFunc()`)

B. It must be declared inside the class using the `friend` keyword

C. It must always be declared as a static function

D. It cannot access private members of the granting class

Correct Answer: B. It must be declared inside the class using the `friend` keyword

21. What is a reference variable in C++?

A. A new independent copy of an existing variable

B. An alias (another name) for an existing variable sharing the same memory location

C. A constant integer variable

D. A pointer that must be dereferenced using `*`

Correct Answer: B. An alias (another name) for an existing variable sharing the same memory location

22. What is the output of the following code snippet?

```
void change(int &x) {
    x = 50;
}

int main() {
    int a = 10;
    change(a);
    cout << a;
    return 0;
}
```

A. 10

B. 50

C. Garbage value

D. Compilation Error

Correct Answer: B. 50

23. Which parameter passing mechanism creates a copy of the argument and does NOT modify the original variable in the caller?

A. Call by Reference

B. Call by Address / Pointer

C. Call by Value

D. Call by Reference-to-const

Correct Answer: C. Call by Value

24. What will be printed by the following Call-by-Value snippet?

```
void fun(int x) {
    x = 100;
}

int main() {
    int a = 10;
    fun(a);
    cout << a;
    return 0;
}
```

A. 100

B. 10

C. 0

D. Compilation Error

Correct Answer: B. 10

25. What is the output of this recursive factorial program when `fact(4)` is executed?

```
int fact(int n) {
    if (n == 0) return 1;
    return n * fact(n - 1);
}

int main() {
    cout << fact(4);
    return 0;
}
```

A. 12

B. 24

C. 16

D. 4

Correct Answer: B. 24

26. What happens if a recursive function lacks a proper base termination condition?

A. It always returns 0 safely

B. It executes only once and stops

C. It leads to infinite recursion causing stack overflow

D. The compiler automatically adds a default base condition

Correct Answer: C. It leads to infinite recursion causing stack overflow

27. Which statement correctly describes default access levels in C++ `struct` vs `class`?

A. `struct` members are private by default, `class` members are public by default

B. `struct` members are public by default, `class` members are private by default

C. Both `struct` and `class` members are private by default

D. Both `struct` and `class` members are public by default

Correct Answer: B. `struct` members are public by default, `class` members are private by default

28. What is the output of the following snippet?

```
int x = 10;
cout << x;
```

A. 0

B. 10

C. x

D. Compilation Error

Correct Answer: B. 10

29. What will be printed by the following code?

```
void show(int x = 5) {
    cout << x;
}

int main() {
    show();
    return 0;
}
```

A. 0

B. 5

C. Garbage value

D. Error: Missing argument

Correct Answer: B. 5

30. What will be the output of executing: `cout << "Hello" << endl << "World";`?

A. `HelloWorld` on a single line

B. `Hello` and `World` printed on separate lines

C. `Hello World` separated by a space

D. Compilation Error

Correct Answer: B. `Hello` and `World` printed on separate lines